
Decision-Level Auditing of Latent Robot World Models

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Abstract

World models are often evaluated by prediction accuracy, but planning depends on the actions their predictions select. We test whether correcting terminal predictions selects physically better actions. On Push-T and Wall, we replace predicted terminal representations with encodings of simulator endpoints reached after executing the same 300 candidate action sequences. Actions, the goal, encoder, and latent scoring objective remain fixed. With released JEPA-WM and DINO-WM checkpoints, predicted scoring outperforms uniform choice on Push-T distance utility and cumulative reward, and on Wall distance utility. Endpoint correction changes 58.4% and 55.9% of valid choices, respectively, but the original tests do not establish additional aggregate physical benefit. Separately, a post-hoc state-distance selector chooses the realized endpoint closest to the goal. It improves distance utility by construction, yet reduces Push-T fixed-target cumulative reward and adds no observed Wall completions. This hindsight diagnostic jointly bypasses the learned representation and latent objective; it is not deployable and cannot isolate an encoder defect. The audit distinguishes useful prediction-based selection, changed decisions, and improved physical outcomes. Its findings concern fixed initial candidates, not full CEM optimization or closed-loop control.

1 Introduction

A world model can predict imperfectly yet rank candidate actions usefully. Conversely, an exact prediction can support a poor decision if the representation and objective used to score it do not match the physical outcome of interest. For example, correcting the predicted position of a robot may change which action has the smallest visual-feature distance to a goal without improving object placement. Prediction accuracy is therefore relevant to planning, but does not by itself specify the value of a prediction correction. Complete-controller success combines prediction, scoring, search, and candidate support; it does not isolate that correction’s contribution.

We ask a controlled question: *if candidate actions, initial state, goal, encoder, and scoring objective stay fixed, does replacing each predicted terminal representation with the representation of what actually happened select a physically better action?* An *endpoint* is the simulator state reached after executing the complete candidate action sequence. We encode its rendered image and proprioceptive observation (the agent’s own state) with the same encoders used by the corresponding planner. Realized-endpoint scoring thus removes terminal prediction error at this interface, but still judges actions through the original latent goal objective. It is not a physical optimum.

We measure physical consequences separately: terminal distance to the supplied goal state, cumulative official coverage reward on Push-T, and completion on Wall. Push-T reward measures block overlap with the environment’s fixed target, not with the replay-generated goal used for terminal distance. A better value on one outcome therefore need not imply improvement on another.

The audit separates four questions. Does predicted scoring contain useful physical selection signal relative to uniform choice? How often does endpoint replacement change the winner? Do those

changes improve physical outcomes? Finally, how much terminal-distance headroom remains when selection bypasses the learned representation and latent objective together? For the last question, we add a post-hoc *retrospective state-distance selector*, which chooses the candidate with the smallest realized simulator-state distance to the goal. It uses outcomes unavailable before execution and is not deployable. It is an upper bound for distance utility on this fixed set, not for every task outcome.

We study Push-T manipulation and Wall navigation using frozen released JEPA-WM and DINO-WM predictors [Terver et al., 2026, Zhou et al., 2025]. Both use pretrained DINOv2 visual features [Oquab et al., 2023], although their preprocessing and proprioceptive encodings differ. We freeze the initial 300 candidates sampled before any cross-entropy-method (CEM) update. Updating their distribution would confound endpoint replacement with candidate generation. The experiment audits the score-to-decision interface on these initial sets; it does not evaluate full CEM or closed-loop model-predictive control (MPC), which would replan after new observations. Each candidate instead receives a complete open-loop replay.

Predicted scoring selects useful actions, and endpoint replacement changes many decisions, but the original tests do not establish additional aggregate physical improvement. The new state selector exposes substantial distance headroom while reducing cumulative Push-T reward and adding no observed Wall completions. The latter comparison is especially limited by sparse successful candidates. These are findings of an audit, not the performance of a proposed repair. Jointly changing representation and objective cannot identify a defect in DINOv2 alone, or distinguish representation limitations from objective choice.

Our contribution is a controlled, candidate-matched evaluation of the physical consequences of terminal prediction correction; a retrospective extension that makes its remaining objective dependence explicit; and a locally verified candidate-level recomputation package prepared for release, with trajectory- and case-level uncertainty. The scope is intentionally narrower than a claim about which world model or planning algorithm is best.

2 Related work

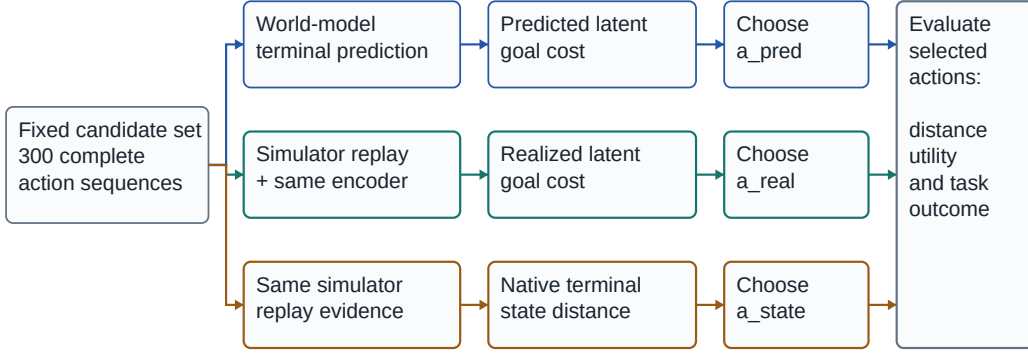
Prediction accuracy and control utility. Lambert et al. [2020] distinguish model-fitting objectives from downstream control performance and study their mismatch in model-based reinforcement learning. Goal-aware prediction [Nair et al., 2020] learns to emphasize task-relevant aspects of future observations. These approaches motivate asking which errors matter to decisions. Our intervention does not train a new loss: it replaces terminal predictions inside an otherwise fixed scoring interface and measures the actions that interface selects.

Value-aware and value-equivalent modeling. Value-aware model learning evaluates model discrepancies through value-function classes rather than generic transition accuracy [Farahmand et al., 2017]. Value equivalence requires models to agree on Bellman updates for specified policy and function classes [Grimm et al., 2020]; it is not generic equality of action rankings. Our finite-candidate audit shares the principle that model quality depends on use, but neither tests Bellman-update equivalence nor establishes a value-equivalence theorem. It directly observes selected actions and their simulator outcomes.

Latent world models and representation-space planning. Latent-dynamics methods support planning without predicting every pixel [Hafner et al., 2019]. DINO-WM predicts pretrained patch features and plans by comparing predicted terminal features with a goal [Zhou et al., 2025]. Terver et al. [2026] study architectural and planning factors in joint-embedding predictive world models, including the imperfect relationship between prediction validation losses and planning success. Predicting a future embedding and measuring its distance to a goal are different uses of a latent space, even when both employ squared error. We retain that goal-scoring interface for the endpoint-correction comparison.

Decision-level evaluation and objective diagnosis. Success rates evaluate the complete interaction of a model, objective, search procedure, and candidate support. Ranking and selected-action diagnostics can separate some of these factors without constituting a new controller. Singh [2026] probe position information and latent-goal geometry in LeWorldModel on TwoRoom, and evaluate changed planning costs under CEM/MPC. Our complementary question concerns fixed candidates:

Same 300 candidates, start and goal. No CEM updates or closed-loop execution.



Predicted -> realized: only the terminal latent input changes.

State arm: bypasses encoder and latent objective; retrospective, not deployable.

Figure 1: Three selectors on the same ordered set of 300 action sequences. Endpoint correction replaces predicted representations with encodings of realized endpoints under the same latent objective. The retrospective state selector additionally bypasses that representation and objective. All selected actions are evaluated on the same native outcomes; only the first selector uses information available before candidate execution.

what changes when only predicted endpoints are replaced, and what remains when a privileged native criterion bypasses the latent interface? We neither isolate the objective causally nor test an objective repair in closed-loop planning.

3 Audit design

3.1 The experiment before the equations

For one initial state and goal, retain 300 candidate action sequences in a fixed order. The world model predicts each terminal representation; the lowest latent goal cost selects a_{pred} . Execute exactly those same sequences from the same simulator initial condition, encode their realized endpoints, and apply exactly the same goal cost to select a_{real} . Evaluate both selected sequences using stored native simulator outcomes. The post-hoc third selector, a_{state} , chooses the smallest stored terminal state distance, then is evaluated on *both* physical outcomes. It is not independently optimized for reward or success.

Figure 1 shows what each comparison changes. Within a model, endpoint replacement holds its encoders, preprocessing, goal, objective, candidate actions, and replay conditions fixed. The state-based comparison changes the selection criterion as well as bypassing the encoders. All comparisons reuse existing candidate outcomes; the diagnostic requires no new training, prediction, or simulation.

3.2 Costs, selectors, and physical outcomes

Let $a_i, i \in \{1, \dots, 300\}$, denote a complete candidate sequence. For the fixed initial observation, the world model produces a terminal latent $\hat{z}_i$; encoding the realized terminal observation produces z_i . The goal encoding is z_g . Each latent has visual and proprioceptive components, denoted by superscripts v and p . The released terminal cost is

$$C(z, z_g) = \frac{\|z^v - z_g^v\|_2^2}{d_v} + 0.1 \frac{\|z^p - z_g^p\|_2^2}{d_p}, \quad (1)$$

$$C_{\text{pred}}(i) = C(\hat{z}_i, z_g), \quad C_{\text{real}}(i) = C(z_i, z_g).$$

where d_v, d_p are the numbers of scalar coordinates in each component. There is no sum of intermediate goal costs. The selectors are

$$a_{\text{pred}} = a_{\arg \min_i} C_{\text{pred}}(i), \quad a_{\text{real}} = a_{\arg \min_i} C_{\text{real}}(i), \quad a_{\text{state}} = a_{\arg \min_i} D(s_i, s_g), \quad (2)$$

with realized terminal state s_i , goal state s_g , and the environment’s native distance D . The new state selector takes the first exact minimum in stored candidate order. Existing selector-validation rules are preserved (Appendix D); no exact minimum ties occur in the valid records.

Both reported outcomes are higher-is-better. Distance utility is $U_{\text{dist}}(a_i) = -D(s_i, s_g)$. The second outcome, $R_{\text{task}}(a_i)$, is cumulative official reward on Push-T and binary completion on Wall. For example, endpoint correction has distance effect

$$\Delta U_{\text{dist}}^{\text{real-pred}} = U_{\text{dist}}(a_{\text{real}}) - U_{\text{dist}}(a_{\text{pred}}). \quad (3)$$

We also report state minus realized and state minus predicted, with the same contrasts for R_{task} . Distance gains for a_{state} are nonnegative by construction; their magnitudes describe available fixed-set headroom. No cumulative-reward guarantee follows on Push-T. On Wall, completion thresholds that same distance, so the state selector also achieves the best available binary completion in each candidate set. Uniform selection has expected outcome equal to the mean over all 300 candidates, so its baseline requires no additional random draw.

4 Experiments and inference

The experiments address four questions: **RQ1**, whether predicted scoring improves physical outcomes over uniform choice; **RQ2**, how often endpoint correction changes the selected sequence; **RQ3**, whether correction improves aggregate physical outcomes; and **RQ4**, what remains under the task-native state criterion. Switching describes decision sensitivity, not an improvement criterion. The primary endpoint-correction contrast includes all retained comparisons rather than conditioning inference on observed switches.

Shared setup. We audit released task-specific JEPA-WM and DINO-WM checkpoints. Each uses a frozen DINOv2 ViT-S/14 visual backbone with 384-dimensional patch features and 224×224 inputs. Proprioceptive encoders and image normalization remain model-specific. Candidates are independent standard-normal samples in the released normalized action coordinates, except the first candidate, fixed at the normalized mean (stored index zero, mathematical index one). Each sequence contains six model actions, each grouping five two-dimensional physical actions: 30 simulator steps in total. Proposal seeds are 1, 2, and 3. No CEM distribution update occurs. Appendix E records transformations, checkpoints, and source bindings. A record is one model’s 300-candidate audit for one initial-state/goal instance and proposal seed. Valid records pass the original artifact, complete-rollout, finite-score/outcome, and selector checks; validity is not physical success.

Push-T. The agent pushes a T-shaped block toward a target configuration, using the released Push-T implementation associated with these checkpoints; Push-T is also used by Chi et al. [2023]. We retain 20 validation trajectories, five predetermined offsets per trajectory, three proposal seeds, and two models: 600 scheduled model-level records. The goal is generated by replaying the corresponding 30-step dataset action segment. The official native distance is raw Euclidean distance over agent position, block position, raw block angle, and agent velocity. It therefore mixes coordinates and units; it is neither pure block-pose error nor task completion. R_{task} sums 30 official per-step rewards, each $\min\{\text{coverage}/0.95, 1\}$, where coverage is the fraction of the environment’s fixed target block area covered by the current block. The target pose is $(256, 256, \pi/4)$ and is not reassigned to the segment’s replay endpoint. Reward lies in $[0, 30]$; it measures coverage over time of that fixed target, whereas distance measures terminal agreement with the segment-specific replay goal.

Analysis counts and missingness. There are 577 valid records and 281 offset–seed cells valid for both models (562 records). The original analysis averages equally over the two models inside each of 20 scheduled trajectory clusters, retaining a fixed denominator of 15 offset–seed cells. Missing or unmatched cells contribute zero *contrast evidence*, not zero physical outcomes. This convention preserves scheduled exposure but attenuates incomplete contributions. This same convention applies to every selector contrast and model split. The reported effect is an observed contrast contribution per scheduled cell, not a mean over observed candidate sets alone. For example, if 14 of 15 scheduled cells are matched and each has a model-averaged contrast of +1, that trajectory contributes 14/15, not 1. The unmatched cell adds no contrast; neither model is assigned a zero physical outcome. The original study also included documented infrastructure recovery after a no-retry rule; Appendix D

reports the deviation, amendment, and an original-completions-only sensitivity that fails the original support rule.

Wall. A point agent navigates around a wall through an opening toward a goal. Thirty fixed initial-state/goal cases, three proposal seeds, and two models give 180 scheduled records. One failed scientific record leaves 179 valid records; inference uses the 29 cases with all six records present (174 records). Native distance is terminal Euclidean position error, and completion means distance strictly below 4.5. Successful candidates occur in only two complete cases. Wall tests whether the correction finding is confined to Push-T’s mixed-unit state metric. It extends the audit to simpler state geometry, not to a strong binary-success benchmark: sparse support restricts completion claims.

Dependence. Direct array checks confirm identical physical candidates and native outcomes for all 281 retained cross-model Push-T pairs and all 89 available Wall pairs. The state-selected action is therefore duplicated across paired model records, not independent evidence. Equal model weighting compares it with two latent selectors; uncertainty resamples whole trajectories or cases, never model rows or candidates. Missing records receive no invented selector or physical outcome.

Original inference. The original inference uses 10,000 cluster-bootstrap draws, paired sign-flip tests, and separate two-outcome Holm families for endpoint correction and predicted-versus-uniform selection within each environment. The original broad-recovery rule required support and stability checks, positive contrasts on both outcomes under raw and bounded proposals, and the original interval and Holm-test conditions (Appendix D). An isolated favorable estimate could not establish that broad claim. We preserve those results unchanged.

Post-hoc diagnostic inference. The state-selector analysis is a new, post-hoc diagnostic using 10,000 whole-cluster percentile-bootstrap draws and no new p -values. A common resampling draw within each environment couples its three contrasts. Consequently, its endpoint-correction point estimates reproduce the original ones but its intervals can differ slightly due to the new shared random seeds. Table 2 reports these new intervals; original intervals and tests remain separately labeled in Appendix B.

5 Results

5.1 Useful selection signal does not imply useful endpoint correction

Predicted scoring carried physical selection signal, but the original tests did not establish an additional aggregate improvement from endpoint correction. On Push-T, predicted-minus-uniform effects were +48.70 in distance utility (95% CI [40.47, 56.43]) and +0.761 in cumulative reward ([0.420, 1.117]), both passing their original Holm-adjusted tests. The corresponding Wall distance effect was +12.363 ([10.900, 13.755]); its small completion effect was not established by the original test. Thus a lack of added benefit from correction should not be confused with an absence of useful signal in the original predicted scores.

For Push-T endpoint correction, the original distance estimate was -5.299 ($[-9.635, -0.706]$, Holm-adjusted $p = 0.0755$) and reward estimate $+0.042$ ($[-0.176, 0.240]$, adjusted $p = 0.701$). The marginal distance interval excludes zero, while the adjusted test does not meet the two-outcome decision threshold; these are different inferential procedures, not interchangeable decisions. Wall effects were -0.254 in distance utility ($[-0.909, 0.364]$) and $+0.029$ in completion ($[0, 0.080]$), each with adjusted $p = 0.923$. These results do not demonstrate equivalence between selectors.

5.2 Endpoint replacement changes many choices

The selectors often disagree even when aggregate physical improvement is not established. Among all valid records, endpoint correction changes 337/577 Push-T choices (58.4%) and 100/179 Wall choices (55.9%). Table 1 distinguishes these descriptive denominators from the records retained for clustered physical comparisons. The state selector usually differs from either latent selector. None of these switch rates by itself indicates whether the new choice is physically preferable. The frequent

Table 1: Descriptive selector disagreements, counts and percentages. “Analysis” retains the original paired-cell or complete-case record set. Records are not independent inferential units.

Environment	Records	Predicted/realized	Realized/state	Predicted/state
Push-T	All valid: 577	337 (58.4%)	525 (91.0%)	515 (89.3%)
	Analysis: 562	331 (58.9%)	511 (90.9%)	502 (89.3%)
Wall	All valid: 179	100 (55.9%)	138 (77.1%)	144 (80.4%)
	Analysis: 174	100 (57.5%)	133 (76.4%)	139 (79.9%)

Table 2: Post-hoc three-selector audit. Effects are first selector minus second, with 95% whole-cluster percentile intervals. Higher is better for both columns. Push-T reward is cumulative coverage; Wall reward is binary completion. Distance units differ across environments. All intervals here use the new common resampling draws, including realized minus predicted; the unchanged original inference appears in Table 3.

Environment	Contrast	ΔU_{dist} [95% CI]	ΔR_{task} [95% CI]
Push-T	Realized — predicted	−5.299 [−9.598, −0.720]	+0.042 [−0.167, 0.242]
	State — realized	+69.814 [62.326, 77.641]	−0.838 [−1.222, −0.460]
	State — predicted	+64.515 [58.162, 71.121]	−0.796 [−1.192, −0.421]
Wall	Realized — predicted	−0.254 [−0.915, 0.363]	+0.029 [0, 0.080]
	State — realized	+5.643 [3.861, 7.664]	0 [0, 0]
	State — predicted	+5.390 [3.902, 7.090]	+0.029 [0, 0.080]

changes nevertheless show that endpoint correction materially affects decisions: the absence of an established aggregate gain is not simply an observation that the selected actions stayed the same.

5.3 State-distance headroom is substantial but objective-dependent

The retrospective state selector improves terminal distance while reducing Push-T cumulative reward and adding no observed Wall completions. Table 2 and Figure 2 report all three contrasts. Push-T effects retain the scheduled-cell denominator described above. Relative to realized-latent selection, Push-T distance utility increases by 69.814 ([62.326, 77.641]), but cumulative reward decreases by 0.838 ([−1.222, −0.460]). Relative to predicted selection, the corresponding effects are +64.515 and −0.796. The selector is a diagnostic, not a proposed controller. It reveals that minimizing replay-goal whole-state terminal distance does not optimize cumulative coverage of the environment’s fixed target. Both the target and temporal/objective definitions differ; the reward decline does not isolate which difference caused the trade-off.

On Wall, state-minus-realized distance utility is +5.643 ([3.861, 7.664]), but the completion contrast is exactly zero in all observed complete cases. Its [0, 0] bootstrap interval reflects resampling those zero contrasts, not an equivalence result or proof of zero population benefit. Only five distinct successful physical candidate instances occur, across four candidate sets in two cases. Predicted, realized, and state selection succeed in 3, 8, and 8 of 174 repeated model records, respectively. Moreover, Wall completion thresholds the same distance minimized by the state selector; it is not independent corroboration of that distance objective. The state selector must complete whenever any candidate can. Its equal completion count therefore means that realized-latent selection already captured every available completion in the retained successful sets. Wall’s zero increment diagnoses limited remaining completion support, not the distance/reward conflict seen on Push-T.

5.4 Model and outcome decompositions qualify the aggregate story

The pooled results conceal different endpoint-correction magnitudes, but both models show the same adverse Push-T reward contrast for state selection. Descriptively, Push-T realized-minus-predicted distance effects are −9.330 for JEPA-WM and −1.268 for DINO-WM; state-minus-realized reward effects are −0.861 and −0.815. On Wall, endpoint-correction distance effects are −0.655 and +0.148, while state-minus-realized distance effects are +4.910 and +6.376. Neither model adds observed completions in that second contrast. Appendix C reports all descriptive model splits; there is no preregistered model-interaction claim and no independent encoder comparison.

Retrospective intervention on fixed candidate sets

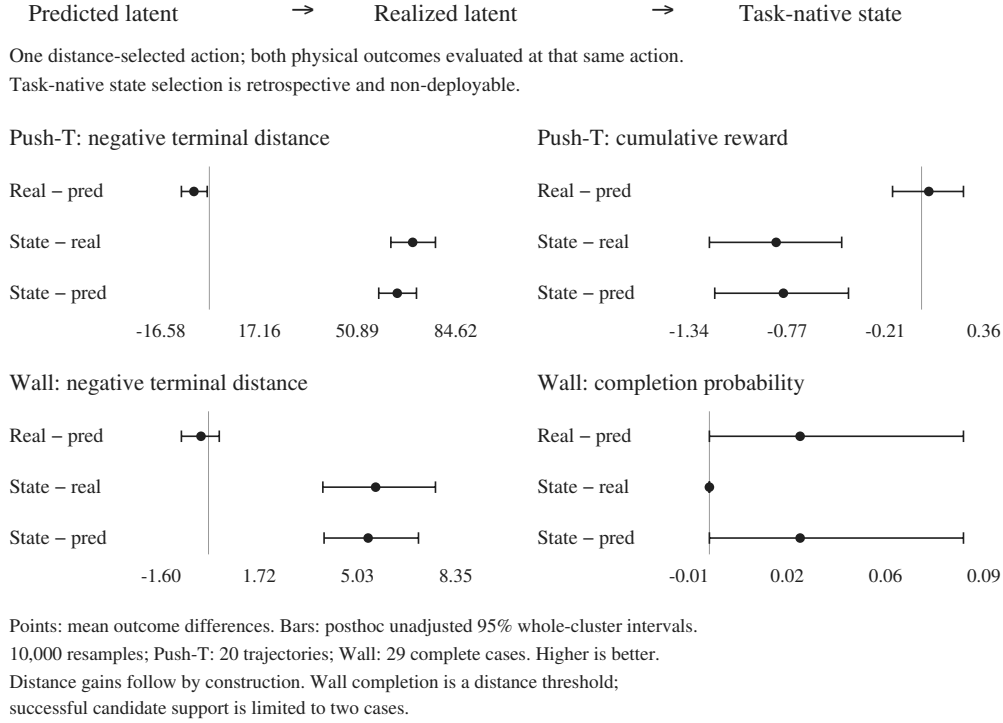


Figure 2: Post-hoc physical effects along the intervention ladder, with whole-trajectory or whole-case 95% bootstrap intervals. The first step corrects terminal predictions under a fixed latent objective. The second jointly bypasses the representation and objective. Distance improvement in the second step is definitional; Push-T reward declines and Wall completion is unchanged in the observed cases. These are diagnostic interventions, not deployable stages of a planning algorithm.

An earlier post-hoc Push-T component analysis further shows why aggregate distance must not be read as block-placement quality. Realized-latent selection improves block-position utility by 1.267 and circular-angle utility by 0.0325 radians, while agent-position and velocity point estimates are adverse. Velocity supplies the largest negative contribution to squared native distance. The component results and their five-component adjustment are retained in Appendix F; they do not identify a causal module. An auxiliary near-tie score predicts selector changes but not established physical harm. Its frozen Wall AUROC, 0.727, is below the simple raw-margin baseline’s 0.735, so it remains a secondary diagnostic rather than a central contribution (Appendix G).

6 Limitations and interpretation

This audit establishes what the retained score-to-decision interfaces do on fixed initial candidate sets. It does not establish full-CEM performance, closed-loop benefit, learned correction quality, real-robot performance, or generalization beyond the two simulation environments. Both predictor families share DINOv2 weights, so they do not supply independent representation-family evidence. Model-specific preprocessing also prevents treating their encoded observations as identical.

Realized-latent selection corrects a terminal input to the objective, not the objective itself. State selection jointly changes the representation and criterion, and requires privileged realized outcomes. Its large distance gains demonstrate available distance headroom, not that DINOv2 is defective, that an objective-only change would recover it, or that a trainable method can attain it. Push-T’s mixed-unit distance and cumulative reward support different conclusions, and its reward target is fixed independently of its replay goal. This prevents attributing the observed reward trade-off purely

to temporal aggregation or to a learned representation defect. Wall’s sparse and threshold-linked completions prevent broad success-rate claims.

The new selector, earlier component analysis, and Push-T near-tie analysis are post-hoc. Whole-cluster resampling respects recorded dependence but cannot remove selection bias from missing data. Push-T’s recovery amendment and fixed-denominator convention, and Wall’s complete-case restriction, remain substantive qualifications. Intervals describe cluster-resampling variation in this selected collection; they do not establish population-wide performance within either task. Marginal intervals are not equivalence intervals; the original sign-flip tests additionally assume null sign symmetry.

7 Conclusion

Useful predicted scoring, sensitivity to prediction correction, and improved physical decisions are distinct properties. Our fixed-candidate audit finds the first two without an established additional aggregate benefit from endpoint correction. Bypassing the latent interface reveals substantial terminal-distance headroom, yet worsens cumulative Push-T reward and adds no observed Wall completions. This objective-dependent result clarifies what the audit measures without proposing a deployable fix or isolating an encoder failure mechanism.

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A Notation and interpretation glossary

Symbol or term	Meaning
a_i	Candidate i : a complete 30-step physical action sequence.
Endpoint s_i	State after executing all actions in candidate i .
$z_i, \hat{z}_i, z_g$	Encoded realized endpoint, predicted terminal latent, encoded goal.
$C_{\text{pred}}, C_{\text{real}}$	The same latent goal cost evaluated on predicted or encoded realized endpoints.
a_{pred}	Candidate selected by predicted latent cost.
a_{real}	Candidate selected by realized-endpoint latent cost; retrospective, not a physical optimum.
a_{state}	Candidate minimizing realized native terminal distance; retrospective and non-deployable.
D, U_{dist}	Native distance and its negative, the higher-is-better distance utility.
R_{task}	Cumulative official Push-T coverage reward or Wall binary completion.
Record	One model, initial-state/goal instance, and proposal seed, with 300 candidates.
Paired cell	One Push-T trajectory offset and seed with valid records for both models.
Cluster	A scheduled Push-T trajectory or a complete Wall case; the resampling unit.
Initial candidates	Samples before any CEM distribution update; historically called an “iteration-0 menu” in code.

B Original inference, unchanged

Table 3 retains the original pooled point estimates, percentile intervals, and Holm-adjusted sign-flip values. Push-T uses all 20 scheduled trajectory contrasts, 10,000 trajectory-bootstrap draws, and enumeration of all 2^{20} sign assignments for its two-sided paired test. Wall uses 29 complete-case contrasts, 10,000 case-bootstrap draws (seed 20260911), and 100,000 random sign assignments with the +1 correction (seed 20260912). Holm adjustment covers two outcomes separately for endpoint correction and predicted-minus-uniform selection within each environment. Push-T’s bounded candidate sensitivity has its own originally declared two-outcome family. The exact test requires sign symmetry of cluster contrasts under the null. No new state-selector comparison has been added to those original families.

Table 3: Unchanged original inference. Intervals are marginal 95% cluster bootstrap intervals; p_H is the original within-family Holm-adjusted value.

Environment	Contrast	Outcome	Effect [95% CI]	p_H
Push-T	Realized – predicted	Distance utility	−5.299 [−9.635, −0.706]	.075539
		Reward	+0.04183 [−0.17569, 0.24024]	.701128
	Predicted – uniform	Distance utility	+48.702 [40.473, 56.434]	7.6294×10^{-6}
		Reward	+0.76099 [0.42033, 1.11714]	.00040245
Wall	Realized – predicted	Distance utility	−0.25365 [−0.90933, 0.36436]	.923091
		Completion	+0.02874 [0, 0.08046]	.923091
	Predicted – uniform	Distance utility	+12.36271 [10.90015, 13.75461]	1.99998×10^{-5}
		Completion	+0.01705 [0, 0.04544]	.501395

The new diagnostic uses seeds 20261002 (Push-T) and 20261003 (Wall), with 10,000 whole-cluster draws shared by its three contrasts and both outcomes within each environment. These intervals quantify the new post-hoc comparisons and do not replace the original test decisions. Recomputing the original results is a required preflight check before generating the new results.

C Model-specific descriptive comparisons

Table 4 uses the same retained cells, cluster definitions, denominators, and common bootstrap draws as the pooled post-hoc analysis. Each model estimate compares a latent selector with the same physical state selector on matched sets. Repeated state-selected outcomes do not provide a second physical experiment. All entries are descriptive; differences between their confidence intervals are not model-interaction tests.

Table 4: Post-hoc model-specific effects with 95% cluster-bootstrap intervals. “Real”, “pred”, and “state” refer to the three defined selectors.

Task	Model	Contrast	ΔU_{dist} [95% CI]	ΔR_{task} [95% CI]
Push-T	JEPA-WM	Real – pred	−9.330 [−14.394, −4.557]	+.081 [−.130, .280]
		State – real	+67.623 [59.887, 76.057]	−.861 [−1.271, −.453]
		State – pred	+58.293 [50.879, 65.943]	−.780 [−1.169, −.407]
	DINO-WM	Real – pred	−1.268 [−7.898, 6.272]	+.003 [−.241, .237]
		State – real	+72.005 [63.922, 80.083]	−.815 [−1.179, −.462]
		State – pred	+70.737 [64.052, 77.512]	−.812 [−1.229, −.420]
Wall	JEPA-WM	Real – pred	−.655 [−1.649, .215]	+.0345 [0, .0920]
		State – real	+4.910 [3.327, 6.785]	0 [0, 0]
		State – pred	+4.255 [3.001, 5.839]	+.0345 [0, .0920]
	DINO-WM	Real – pred	+.148 [−.724, .923]	+.0230 [0, .0690]
		State – real	+6.376 [4.222, 8.825]	0 [0, 0]
		State – pred	+6.524 [4.551, 8.761]	+.0230 [0, .0690]

D Sampling, missingness, and the recovery amendment

Push-T selection and support. The manifest selects validation trajectory indices 0–20 except 7, for 20 trajectories, without consulting audit outcomes. For a trajectory of length L , its five offsets are $\lfloor k(L - 31)/4 \rfloor$, $k = 0, \dots, 4$. Only recorded sequence lengths select the offsets. Expert block-state motion labels a secondary phase proxy but does not select or exclude records. Each offset is paired with proposal seeds 1, 2, and 3 and both models. The resulting 100 distinct initial-state/goal segments produce 300 offset–seed cells and 600 scheduled model records. The manifest’s historical `segment_count=300` includes proposal seeds.

For an outcome Y (either U_{dist} or R_{task}), let δ_{tcm} be a specified selector contrast for trajectory t , cell c , and model m . The cell indicator I_{tc} equals one only when both model records are valid under the original rules. The pooled trajectory contrast is

$$\bar{\delta}_t = \frac{1}{2} \sum_{m=1}^2 \frac{1}{15} \sum_{c=1}^{15} I_{tc} \delta_{tcm}, \quad \hat{\Delta} = \frac{1}{20} \sum_{t=1}^{20} \bar{\delta}_t. \quad (4)$$

Missing-cell terms are defined as zero *contrast contributions*; the formula does not assign a physical outcome to a missing action. The two-model mask and fixed 15-cell denominator are preserved for every new contrast and model split. Observed-cell means would answer a different question. The original support requirement is at least 540 matched model records and at least 18 trajectories with 12 or more matched cells. The retained study has 562 matched records and 19 such trajectories, so it meets that requirement; all 20 scheduled trajectories still enter Eq. 4. Support gates require coverage of the scheduled experiment, not a positive physical effect or a guarantee of adequate power. The original broad-recovery rule additionally required passed integrity and repeat-stability checks; positive pooled means and lower 95% bounds, with Holm-adjusted $p < .05$, for both outcomes under raw and bounded candidates; positive model-specific lower bounds for both outcomes in both candidate sets; and no conflicting pooled signs. Failure of any condition blocked the broad claim. Sampling, outcomes, aggregation, tests, and this rule were recorded in the original locked protocol; recovery was a later amendment, and the auxiliary and new state-selector analyses are separately labeled post-hoc.

Infrastructure recovery. The Push-T protocol was recorded as locked on August 30, 2026 and prohibited retries. Its original 600-record attempt ledger ended with 420 completed and 180 failed

records. A fatal CUDA error had contaminated the persistent worker, causing later untouched records to receive the same infrastructure error. The recovery amendment mechanically selected 169 failed records with that exact error message for at most one attempt each, in isolated processes. The remaining failure classes were not eligible. The amendment states that eligibility was fixed before inspecting scientific outcomes, preserving candidate specifications and analysis rules, but this remains a deviation from the original no-retry rule. Original failures remain in the ledger; recovery neither deletes them nor makes them original completions. The no-retry rule restricted opportunities to change retained evidence through selective repeated attempts; the amendment does not retroactively satisfy it. The recovery-augmented analysis has 577 valid and 23 invalid records, with 15 valid records unmatched across models. Removing every recovery-sourced artifact leaves 408 matched records and only 14 trajectories meeting the original eligibility threshold: the original-only subset fails the original support requirement. Its distance and reward effects are -5.39 ($[-9.32, -1.40]$) and -0.019 ($[-0.212, 0.160]$), respectively. These are exploratory sensitivity estimates, not a re-stored confirmatory analysis or evidence that missingness is ignorable. The 15 observed trajectory groups used by a separate diagnostic are not 15 eligible primary clusters.

Earlier protocol versions are not scientific inputs to this analysis: v1.2 failed its prescribed RGB/serialization integrity check, and v1.3 failed before usable measurements because a replay-encoding helper lacked its required model argument. The v1.4 source and evidence are separately bound; no earlier-version result is reused. The retained protocol, amendment, and ledgers make this history inspectable but do not constitute an independently registered timestamp for the declared outcome-blind sequence.

Wall selection and missingness. Case seeds 2026091100–2026091129 generate the 30 initial-state/goal cases. Engineering seeds 2026091001 and 2026091002 are excluded from the study. The wall location is 32 and door location 30 in the underlying 65-pixel environment. Each case contains seeds 1–3 and both models. One JEPA-WM record, case wall_028, seed 1, failed scientific validation and was not retried. Its entire case is excluded from the complete-case physical analysis. The remaining five valid records contribute only to all-valid descriptions. Case contrasts average the six model–seed records equally; the 29 case means are the inferential units. This restriction does not rule out difficulty-dependent missingness. The retained failure ledger labels this failure `scientific_or_child` but supplies no error message, so its specific failed criterion is not established by that ledger.

Integrity and ties. Original Push-T validity required finite production arrays, decomposition residuals at most 10^{-8} , and score reproduction within 2×10^{-5} with zero relative tolerance. Re-encoding serialized observations at chunk sizes eight and four had to preserve scores within that tolerance and the latent minimum tie sets exactly. Original repeat and provenance checks also applied. Execution completion alone therefore did not imply scientific validity; a valid record still needed its partner for paired analysis. The new preflight checks cover all 780 unique scheduled records: 756 valid and 24 invalid across both environments. They verify candidate counts, selectors, stored physical arrays, cross-model agreement, and the original headline analysis before new computation. Push-T’s original latent selectors require a unique minimum under absolute tolerance 10^{-10} and relative tolerance 10^{-8} ; otherwise that record is invalid. Wall’s latent selectors use the first exact minimum. The new state selector also uses the first exact minimum. No exact minimum ties or duplicate physical sequences occur in the retained valid candidate sets; synthetic tests cover deterministic tie handling. Cross-model identity is established by arrays, not assumed from shared random-seed labels.

E Model and environment configuration

Visual encoding. Both models in both tasks load `dinov2_vits14` from `dinov2_vits14_pretrain.pth`: a ViT-S/14 with 384-dimensional patch features. Raw RGB is divided by 255 and resized to 224×224 using bilinear interpolation with `align_corners=False`. The audit loaders fix spatial scale and aspect-ratio augmentation ranges to (1,1) and disable flips, motion augmentation, automatic augmentation, and random erasing. DINO-WM then uses per-channel mean and standard deviation (0.5, 0.5, 0.5). JEPA-WM uses mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). Sharing backbone weights therefore does not mean that the models receive identically normalized inputs. Each model’s preprocessing is fixed between its predicted and realized costs.

Proprioception and actions. Push-T proprioception contains the agent’s two-dimensional position and velocity (four coordinates), not block pose. Wall proprioception is the two-dimensional agent position. After the released coordinate normalization, the frozen proprioceptive encoder is a kernel-one, stride-one convolution, equivalent to a per-time-step affine map. Its output dimension is 20 for DINO-WM on Push-T and 10 on Wall, versus 16 for JEPA-WM on both tasks. Table 5 gives the environment normalization constants; the operational meaning of action normalization is $a_{\text{physical}} = \sigma_a \odot a_{\text{normalized}} + \mu_a$. Consequently, candidate zero is a normalized-mean sequence, not necessarily a physical no-op. Gaussian proposals are unbounded in normalized coordinates; the bounded sensitivity is defined separately, not silently substituted.

Table 5: Released environment normalization constants, rounded here for readability. Full-precision values remain in the bound dataset registry.

Task	Quantity	Values
Push-T	Action mean	(−0.0087, 0.0068)
	Action standard deviation	(0.2019, 0.2002)
	Proprioception mean	(236.61549, 264.56741, −2.93032, 2.54308)
	Proprioception standard deviation	(101.12020, 87.01120, 74.84556, 74.14009)
Wall	Action mean	(−0.00556462, 0.01319782)
	Action standard deviation	(0.75233287, 0.74668980)
	Proprioception mean	(31.89451, 31.81402)
	Proprioception standard deviation	(17.66352, 17.27517)

The model horizon is six, action dimension ten, physical action dimension two, and frame grouping five, giving 30 executed actions with `action_skip=1`. Initial image and proprioceptive history lengths are one; rollout context is two. The objective is terminal visual MSE plus 0.1 proprioceptive MSE, with `sum_all_diffs=False`. Push-T samples with a PyTorch generator on the initial latent’s device; Wall uses a CPU generator with shape [6, 300, 10] before permutation. Both use standard-normal proposals, but their random streams should not be assumed identical across environments.

Dataset and goals. Push-T uses `data/pusht_noise/val` in the retained official source tree. The simulator starts from `states[offset]` with the recorded shape, and replays the 30 recorded relative actions using environment seed 1 to generate the goal. The goal is not merely a frame extracted from the dataset video. Wall cases are generated from the fixed random-state seeds described above rather than validation-trajectory offsets. The capture routine `run_case` in `experiments/wall_decision_audit.py` calls the released `WallEnvWrapper.sample_random_init_goal_states`, which delegates to `DotWall.generate_random_state` in `evals/simu_env_planning/envs/wall_env/envs/wall.py` under the bound Wall configuration. It samples positions uniformly on opposite sides of the wall within padded boundaries, reversing travel direction with probability one half. These routines, not the seed alone, define case generation.

Source and checkpoint bindings. The official JEPA-WM source commit is `13cf1d9c7e476f53c17714d2e0f1dc239a883ce0`; the DINOv2 source commit is `b194f00db6136677fc8a4cc2ef2168f7699dfba2`. The recorded Wall model-repository revision is `9b9c41ef249466630dbf1a20e78391865d07b3b9`. The checkpoint filenames and SHA-256 digests are:

- `dino_wm_pusht.pth.tar`:
8ec9cb05f22812d7f12e3c216b0637f41641055c0653e503e2746edb981b550f.
- `jepa_wm_pusht.pth.tar`:
9beca3eafe0739c3b3adb5d734fa435ccbda0fea8a65d53d4cccec176aaaa0eb.
- `dino_wm_wall.pth.tar`:
ff170be5aec9249768be4a220d600b8f00a8589b2a78982ecf9273809f2767df.
- `jepa_wm_wall.pth.tar`:
8efb0623cfba1cb3ca210de26f7579c83dd24936635f11989c515afcb23bea1e.

The DINOv2 weight-file digest is `b938bf1bc15cd2ec0feacfe3a1bb553fe8ea9ca46a7e1d8d00217f29aef60cd9`; the recorded loaded state-dictionary digest

is 5bb315664719c85935c4b67a0f8faef9d01b7dfb57ffbecef1c062f7b4e4aa35. The Push-T dataset archive digest is 442f5dee246edf670964ed7bdec248683cd6d00580fa0e4d458abb53f92da08.

These facts are bound by experiments/official_pusht_v14_manifest.json (selection, protocol, source_bundle); output/source/physical_ai_supplement/official_pusht_v14_certificate_issuance.json (model, encoder, and data bindings); experiments/wall_audit_20260910/protocol.json; experiments/wall_audit_20260910/execution_bindings.json; and output/wall_recompute_20260910_v3/metadata/provenance.json. The official planning configurations lie under configs/evals/simu_env_planning/, with task directories pt and wall, model directories dino-wm and jepa-wm, and names ending L2_cem_sourcedset_H6_nas6_ctxt2_r224_alpha0.1_ep96_decode.yaml for Push-T or the analogous sourcerandstate name for Wall. Normalization constants are in app/plan_common/datasets/__init__.py of the bound source tree. Checkpoint training epochs, training seeds, and complete training-data provenance are not established by these retained audit bindings; we do not infer them from filenames.

F Earlier sensitivities and exact score accounting

These analyses predate the state-selector extension and retain their original status and numerical summaries. They are not new experiments for this revision. The bounded-candidate and original-completions-only sensitivities concern endpoint correction, not the new state-selector contrasts. We have not reported an observed-cell-denominator analysis or an original-completions-only state-selector analysis. Their absence leaves the corresponding robustness questions unresolved; the existing sensitivities cannot establish that missingness is ignorable for any contrast.

Bounded candidates. Restrict each set to sequences with maximum absolute normalized action coordinate at most three. The original support requirement is at least 200 candidates and valid untied latent selectors. The realized-minus-predicted effects are -5.80 in distance utility (95% CI $[-11.00, -0.21]$, Holm-adjusted $p = .111$) and $+0.115$ in reward ($[-0.093, 0.314]$, $p = .296$). The raw analysis remains primary; this sensitivity does not alter the new state-selector definition or its candidate set.

Physical components. The component analysis was post-hoc, specified before inspecting component outcomes. It uses the same 562 matched records, 281 cells, 20 trajectories, and fixed denominators. Position and velocity utilities are negative Euclidean distances to their goal components; angle utility is negative circular absolute error, unlike the raw angle coordinate in native distance. Terminal coverage reward is the last official reward, not the raw overlap fraction reported separately by the environment. Holm adjustment covers these five declared component tests.

Table 6: Retained post-hoc Push-T component effects, realized minus predicted. Intervals are original marginal 95% trajectory-bootstrap intervals.

Component utility	Effect	95% CI	p_H
Agent position	-1.817	$[-4.564, .971]$	.232
Block position	$+1.267$	$[-.480, 2.176]$	.0203
Circular block angle	$+.0325$	$[-.0197, .0471]$	.000210
Agent velocity	-5.809	$[-10.523, -1.187]$	.0957
Terminal coverage	$+.00901$	$[-.00035, .01696]$	.110

For exact native squared-distance accounting, the mean predicted-selection minus realized-selection contributions are -477.2 from agent position, $+79.2$ from block position, $+0.226$ from raw angle, and -1497.0 from agent velocity. Positive contributions mean realized selection reduces that squared error. The maximum rowwise identity residual is 3.64×10^{-11} . These mixed-unit squared contributions are not additive contributions to mean unsquared distance utility, and they change under coordinate rescaling.

Stored-latent interpolation. For each candidate, the earlier post-hoc sensitivity scores $(1 - \lambda)\hat{z}_i + \lambda z_i$, for $\lambda = 0, .1, \dots, 1$, with all actions, goals, costs, and physical outcomes otherwise unchanged. At $\lambda = .5$, 38.4% of matched-record choices change and the distance effect is -2.24 ; at $\lambda = 1$, these values are 58.9% and -5.30 . All ten nonzero- λ pooled distance point estimates are negative. These aggregate points are not independent replicates, interpolated latents need not represent realizable states, and no trainable correction is evaluated.

Score accounting is an identity, not a causal explanation. For either modality, write its prediction error as $e_i = \hat{z}_i - z_i$, with dimension d . Then

$$\frac{\|\hat{z}_i - z_g\|_2^2 - \|z_i - z_g\|_2^2}{d} = \frac{2(z_i - z_g)^\top e_i}{d} + \frac{\|e_i\|_2^2}{d}. \quad (5)$$

Summing the two modalities with weights one and 0.1 reconstructs $C_{\text{pred}}(i) - C_{\text{real}}(i)$ exactly; subtracting two candidates' identities reconstructs their margin change. The maximum residual in retained Push-T records is 6.66×10^{-16} . Directional terms dominate magnitude terms in 266 of 337 selector switches (78.9%): the absolute directional margin between the predicted and realized winners exceeds the absolute magnitude margin after the visual and 0.1 proprioceptive contributions are combined. This is a record-level fraction, not cluster-level inference. This explains algebraic score movement, not its physical desirability or a causal failure mechanism. The trajectory-weighted mean candidate-set Spearman correlation between the two costs is .858 (95% CI [.839, .875]); the mean per-set median absolute cost difference is .0505, with mean ratio .339 to predicted-cost IQR.

Ranking summaries. Spearman correlates negative cost with higher-is-better outcome, using average ranks for exact ties. Pairwise accuracy counts whether lower cost favors the higher outcome, excluding pairs tied on either quantity. Per-record summaries use the original paired-cell, equal-model, trajectory aggregation. On bounded candidate support, the earlier descriptive distance-utility Spearman correlations are .367 for predicted and .345 for realized scoring; pairwise ranking accuracies are .599 and .591. Their normalized regrets are .250 and .274. For cumulative reward, the corresponding pairs are .121/.124, .469/.471, and .372/.359. For a higher-is-better outcome Y , normalized regret of selection a is $(\max_i Y(a_i) - Y(a)) / (\max_i Y(a_i) - \min_i Y(a_i))$ when its denominator is positive and the selector is defined. These summaries are descriptive and do not establish ranking superiority.

G Secondary selector-instability diagnostic

The earlier near-tie diagnostic targets whether a_{pred} and a_{real} differ, not whether either fails physically. After deduplicating physical actions and retaining the lowest predicted cost per action, let $c_{(1)}$ and $c_{(2)}$ be the two lowest predicted costs. The ambiguity score is

$$-\log_{10} \left[\max \left\{ \frac{c_{(2)} - c_{(1)}}{\max\{\text{IQR}(C_{\text{pred}}), 10^{-12}\}}, 10^{-12} \right\} \right]. \quad (6)$$

The Push-T analysis was exploratory. Leave-one-trajectory-out logistic models include model identity, with model-specific top-quartile thresholds learned on the other 19 trajectories. Bootstrap intervals resample saved out-of-fold predictions without refitting the complete pipeline, and do not account for choosing this diagnostic after the primary result.

On 562 matched records, unfitted normalized-ambiguity-score AUROC is .714 ([.665, .762]), compared with .706 for the fitted probability. The flagged group covers 25.3% of records and switches 82.4% of the time versus 51.0% outside it. However, flagged-minus-unflagged normalized regret of a_{pred} is $-.007$ for distance utility ([$-.039, .026$]) and $-.056$ for reward ([$-.143, .026$]), neither establishing greater physical regret.

Before the secondary Wall extension, model thresholds were frozen at 1.2899081787 for JEPA-WM and 1.4614315032 for DINO-WM, without Wall retuning. Across all 179 valid records, the normalized score has AUROC .727 versus .735 for raw margin. These descriptive estimates provide no normalization advantage. The signed native gap is $\min_i D(s_i, s_g) - D(s_{a_{\text{pred}}}, s_g)$; larger values mean less distance regret. Its flagged-minus-unflagged effect is $+.393$ ([$-1.704, 2.627$]), using the 19 complete cases containing both groups. Thus prospective freezing limits retuning but does not establish physical-risk calibration, verification benefit, or an improvement over the simpler baseline.

H Artifacts, recomputation, and scope

The new analysis reads retained candidate scores, actions, terminal states where available, native distances, and task outcomes. It verifies candidate identity across models and reproduces the original effects, intervals, tests, and families before computing the state-selector extension. Its numerical review independently matched all 36 pooled/model-specific summaries and intervals, valid-record selector outcomes, and cluster arrays. Original source artifacts and result files are not overwritten. These revision materials are retained locally for author release; this work does not claim that the new package has already been publicly uploaded.

The analysis artifact directory `output/task_native_selector_20261002/` contains the checked specification and implementation reviews, `summary.csv`, `row_effects.csv`, `cluster_effects.csv`, `selection_disagreements.csv`, `original_results.json`, `physical_support.json`, `RESULTS_INTERPRETATION.md`, reproduction instructions, tests, and a portable reproduction bundle. `SCIENTIFIC_FREEZE.md` records the pre-manuscript numerical freeze. The numerical summary SHA-256 is `3eb5cb00fd7be3f9e01a6f1e6277b00302d0e1549c7d8c8845fba59570bb09c6`. The reviewed result SHA-256 is `f679e348860c2e4db3af4eb98c506b29532127482c2fd60135d8d142801ca8d9`. Repository and portable relative commands are provided in the respective `README.md` files. The portable package was checked from a relocated directory with Python’s isolated mode and an existing NumPy installation; all declared numerical outputs matched exactly. This is not a fresh dependency installation or simulator rerun. Python and NumPy suffice for the new CPU analysis; it does not run models or environments.

Stored native distances are authoritative for retrospective selection. The compact Push-T export does not contain full initial-state and raw-image evidence, and exported Wall terminal states have rounded float32 precision; recomputing native distances from those rounded arrays would change the numerical object being audited. The retained distances and reward arrays are sufficient for the reported retrospective comparisons, not for complete regeneration of the original simulator runs.

The earlier compact Wall bundle contains 179 valid model records with 300 candidates each: 53,700 model–candidate records. Repeated physical sets across models remain separate prediction records, not independent physical trials. Its 180-record manifest and failure ledger preserve the failed row. The Push-T manifest, preregistration, recovery amendment, issuance and attempt ledgers, analysis outputs, and source bindings are retained separately. The artifacts support analysis recomputation, not reproduction of model training or a claim that every raw observation is included in one compact export.

Tool assistance and responsibility. LLM agents assisted implementation, analysis scripting, verification, and drafting. Numerical claims were checked against retained artifacts and an independent recomputation. The authors remain responsible for study design, scientific claims, and presentation; agent assessments are not empirical evidence. These simulator audits provide no deployment or robot-safety guarantee.